FUNCTIONAL REQUIREMENTS & VALIDATION REPORT

TI Piccolo-Based Dual-Motor Dynamometer Testbed

PUBLIC ENGINEERING RELEASE v1.4

Document ID	HES-DYNO-FRD-001
Prepared by	Herder Elektronische Systemen
System basis	TI C2000 Piccolo F28069M dual-PMSM dynamometer testbed
Purpose	Define the public system boundary, functional and quality requirements, verification methods, objective evidence, validation status, and release limitations.

DOCUMENT POSITIONING

Independent engineering documentation for a platform built with Texas Instruments development hardware. This document is not authored, approved, or sponsored by Texas Instruments. Product names and trademarks remain the property of their owners.

1. Executive Summary

The platform is a dual-PMSM dynamometer centered on a TI C2000 Piccolo F28069M real-time controller, two DRV8305 inverter stages, quadrature encoder feedback, and a host-side test and analysis workflow. The inherited platform contained usable hardware and control assets, but the documented motor-phase and encoder connections did not match the physical bench. The electrical interface was corrected, dual-machine control was restored, host-target pacing was stabilized, an efficiency-map workflow was built, and the communications path was extended from SCI-based operation to CAN-controlled real-time testing.

This FRD converts the public evidence set into controlled, testable requirements. Each normative requirement is uniquely identified, written as a single “shall” statement, assigned a verification method, and traced to retained evidence. The structure is aligned with ISO/IEC/IEEE 29148 requirements-engineering principles and NASA requirements/verification guidance; it is not a claim of formal certification to either framework.

Demonstrated capability	Public evidence
Dual-machine operation	Corrected phase and QEP interface; stable commanded drive/load behavior.
Operating envelope represented in data	0-4000 rpm and 0-0.25 N m processing range.
Measured efficiency characterization	Raw-signal review, validity masks, measured-bin maps, sample counts, and interpolation views.
Real-time communications	CAN receive ID 0x100 and transmit ID 0x101 in retained generated-code evidence; four packed 16-bit command values and telemetry organization.
Drive-cycle workflow	Compressed FTP-style command profile and load-command generation for bench demonstration.

2. Document Scope and Conventions

2.1 Public system boundary

- Included: system architecture, interface definitions, actual editable Simulink target/host models, custom F28069M ADC configuration source, selected MATLAB utilities, sanitized plots, vendor schematic references, public BOM, requirements, and verification evidence.
- Excluded: generated code trees, compiled firmware and binaries, vendor device-support source, private tuning history, raw unpublished data, editable PCB/manufacturing packages, and internal development records.
- Third-party boundary: TI boards and documentation, MathWorks software, Speedgoat interfaces, motors, and inherited fixtures remain third-party or inherited elements.

2.2 Requirement language and status

Term	Meaning
Shall	Mandatory behavior or constraint for the public baseline.
Should	Recommended practice or future enhancement; not mandatory.
Verified	Objective evidence supports the requirement for the reviewed baseline.
Partially verified	Some evidence exists, but an identified condition remains open.
Not verified	No adequate objective evidence is included in the public release.

2.3 Verification methods

Code	Method	Definition
I	Inspection	Review of drawings, files, configuration, labels, or retained source artifacts.

Code	Method	Definition
A	Analysis	Evaluation by calculation, data processing, or engineering reasoning.
D	Demonstration	Observed operation without exhaustive quantitative acceptance limits.
T	Test	Controlled execution with defined input, measured output, and acceptance criterion.

3. System Context and Architecture

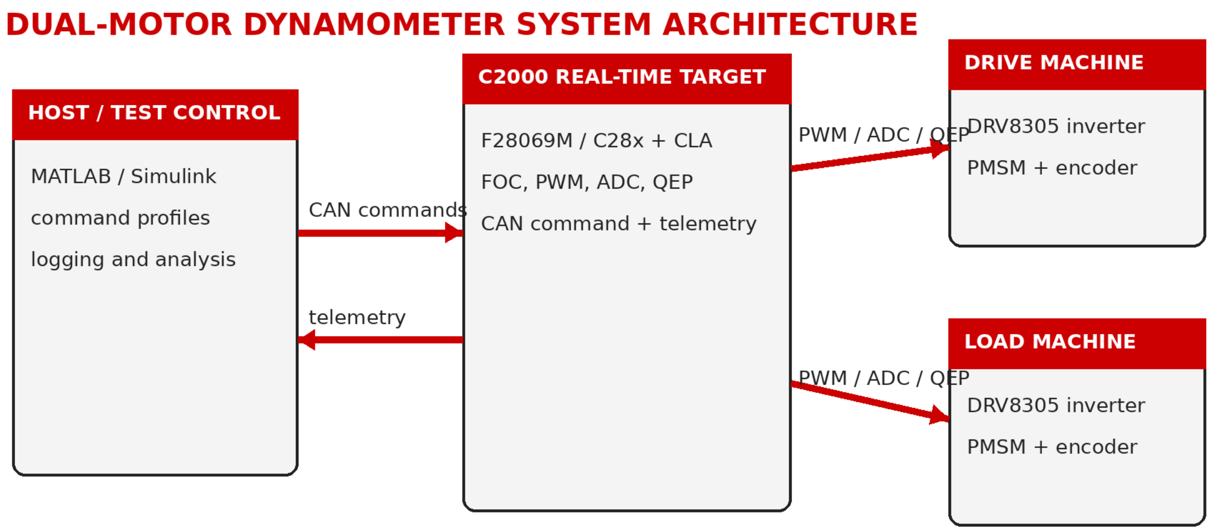

Figure 1. Public system architecture and responsibility boundary.

Time-critical motor control remains on the F28069M target. The host creates command profiles, supervises tests, logs selected channels, and performs analysis. The CAN link replaces dependence on host GUI refresh timing for the command/telemetry transport. Each inverter controls one PMSM; each encoder is assigned to the corresponding QEP input after physical verification.

3.1 Primary interfaces

Interface	Public definition	Boundary
DC power	Nominal 24 V bench supply to both inverter stages.	Exact protection and supply design remain bench-specific.
Motor phases	Three-phase outputs from each DRV8305 inverter to its assigned PMSM.	Phase order verified physically; no universal harness pinout is claimed.
Encoder feedback	Quadrature A/B feedback to F28069M QEP peripherals.	Exact connectorization is platform-specific.
CAN	Differential CANH/CANL with common reference and termination.	Public baseline identifies command 0x100 and response 0x101.
Host data	Selected signals exported for synchronized analysis.	Complete private signal dictionary is excluded.

4. Functional Requirements

ID	Requirement	V-method	Status	Evidence
SYS-001	The system shall provide	D/T	Verified	Operational dual-motor bench

ID	Requirement	V-method	Status	Evidence
	independently commanded drive-machine and load-machine control paths.			evidence.
SYS-002	The system shall maintain an explicit mapping between each inverter, motor, and encoder feedback channel.	I/T	Verified	Corrected phase and QEP assignments.
CTL-001	The real-time target shall execute motor-control timing independently of host visualization refresh timing.	I/D	Verified	Target FOC/PWM/ADC/QEP architecture.
CTL-002	The target shall provide closed-loop speed regulation for the commanded drive machine within the demonstrated bench envelope.	T	Verified	Successful speed-command tests and efficiency-map runs.
CTL-003	The target shall provide a controllable load-machine q-axis current or equivalent torque-producing command.	T	Verified	Fixed-dwell and drive-cycle load profiles.
CTL-004	The control path shall preserve enable, limit, and command-interpretation logic rather than bypassing the established motor-control state path.	I/D	Verified	Retained host/target model behavior and command routing.
COM-001	The target shall receive a standard CAN command frame with identifier 0x100.	I/D	Verified	Generated target code mailbox configuration.
COM-002	The target shall transmit a standard CAN response frame with identifier 0x101.	I/D	Verified	Generated target code mailbox configuration.
COM-003	A command frame shall transport four packed 16-bit values using a documented scaling convention.	I/D	Verified	Byte pack/unpack and host command evidence.
COM-004	The public interface description shall distinguish command transport from measurement telemetry.	I	Verified	Interface summary and FRD architecture.
DAQ-001	The acquisition workflow shall align selected signals to a common analysis interval and time base before derived calculations.	I/A	Verified	Interpolation and crop logic in processing source.
DAQ-002	The processing workflow shall reject non-finite and physically implausible samples before map generation.	I/A	Verified	Validity masks and filtering plots.
DAQ-003	The workflow shall retain sample-count evidence for each measured operating bin.	A	Verified	Sample-count map.
CHR-001	The efficiency workflow shall support an analysis grid spanning 0-4000 rpm and 0-0.25 N m.	I/A	Verified	Processing edges and published figures.
CHR-002	Electrical input power shall be derived from the selected q-axis voltage/current representation used by the retained workflow.	I/A	Verified	Processing script and report method.
CHR-003	Mechanical output power	I/A	Verified	Processing script and report

ID	Requirement	V-method	Status	Evidence
	shall be derived from measured or estimated torque and angular speed.			method.
CHR-004	Measured-bin results shall remain distinguishable from interpolated visualization products.	I	Verified	Separate measured and interpolated maps.
DRV-001	The host workflow shall generate a bounded, compressed FTP-style speed command suitable for a short bench demonstration.	I/D	Verified	Public drive-cycle setup script.
SAF-001	The release shall not characterize the platform as production-safe or safety-certified.	I	Verified	NOTICE and report limitations.
REL-001	The public package shall exclude complete private control models, generated code, compiled firmware, and editable manufacturing sources.	I	Verified	Final package inspection.
SYS-021	The target implementation shall configure the additional simultaneous ADC input pairs required by the public instrumentation path using project-specific F28069M SOC configuration source.	I/T	Verified	F28069M_ADC_Custom_Config.c; SOC12/13 and SOC14/15 configuration.

5. Performance and Quality Requirements

ID	Requirement	Method	Status
PER-001	The public characterization workflow shall represent speed commands from 0 through 4000 rpm.	I/A	Verified
PER-002	The public characterization workflow shall represent torque values from 0 through 0.25 N m.	I/A	Verified
PER-003	Efficiency-map acceptance shall require electrical power greater than 10 W, mechanical power greater than 3 W, and torque greater than 0.045 N m.	I/A	Verified
TIM-001	Host-target execution shall be paced sufficiently to avoid uncontrolled host run-ahead during real-time tests.	D	Verified
USA-001	Public scripts shall expose configuration parameters and generate labeled plots suitable for engineering review.	I	Verified
MAI-001	The public repository shall separate firmware evidence, host utilities, hardware references, results, media, and documentation.	I	Verified
TRC-001	Every FRD requirement shall have a unique identifier, verification	I	Verified

ID	Requirement	Method	Status
	method, status, and evidence reference.		
SEC-001	The public release shall contain no credentials, customer data, university-only records, or internal planning notes.	I	Verified
PER-009	The added ADC pairs shall use PWM-synchronized triggering and explicit end-of-conversion interrupt routing so acquisition remains aligned with target execution.	I	Verified

6. Verification Evidence

6.1 Electrical and control recovery

The working baseline was established by reconciling the physical bench with the inherited connection documentation. Three-phase order and encoder/QEP assignments were corrected, followed by staged single-machine and dual-machine operation. This evidence supports the interface-mapping and closed-loop-control requirements, but it does not establish production wiring compliance or immunity to incorrect reconnection.

6.2 CAN integration evidence

Retained generated-code evidence configures mailbox 0 to receive identifier 0x100 and mailbox 1 to transmit identifier 0x101. The transport packs four 16-bit command values per eight-byte frame. The customer-facing release describes the transport and scaling boundary while while excluding the complete generated-code tree and private target models.

6.3 Efficiency-map evidence

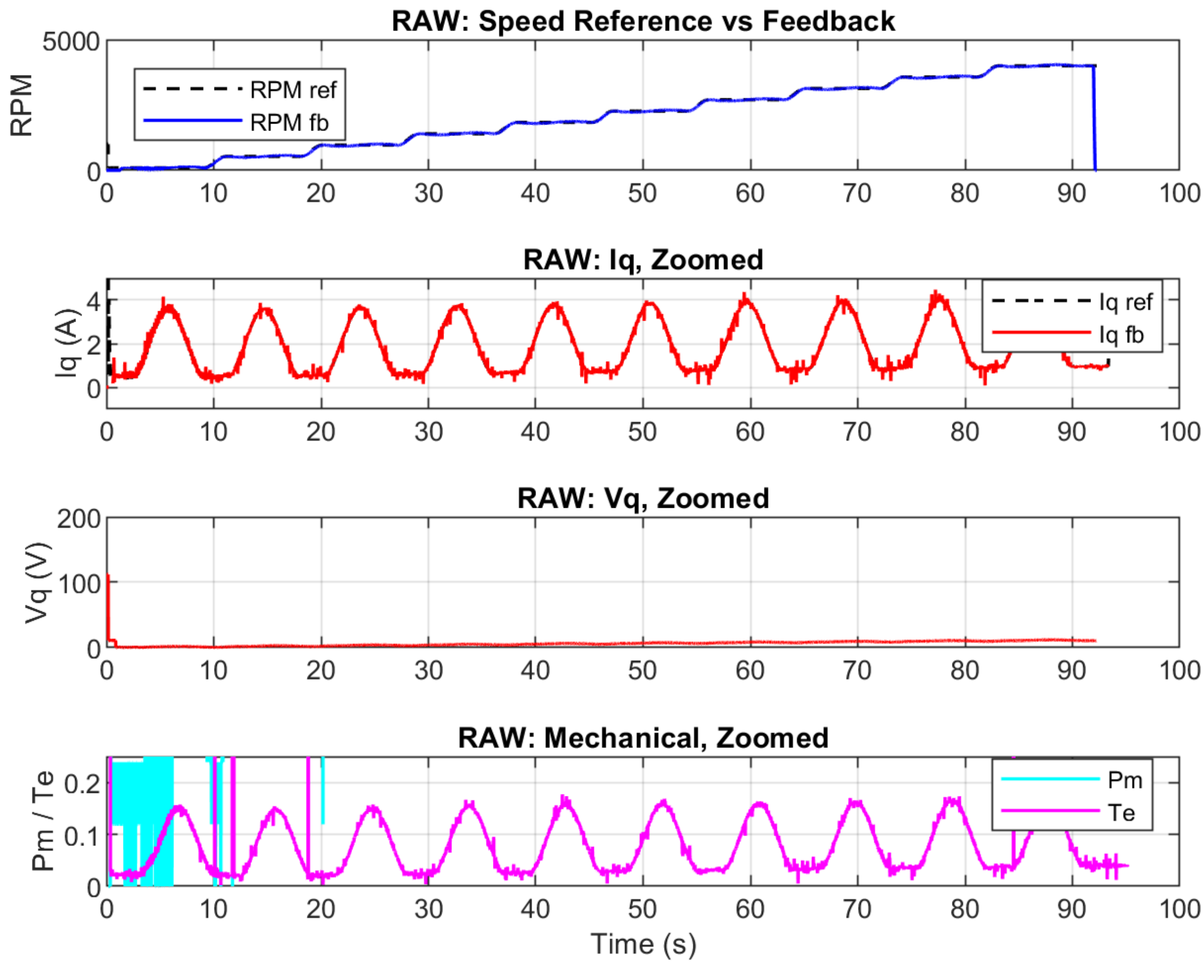

Figure 2. Raw selected signals before validity screening.

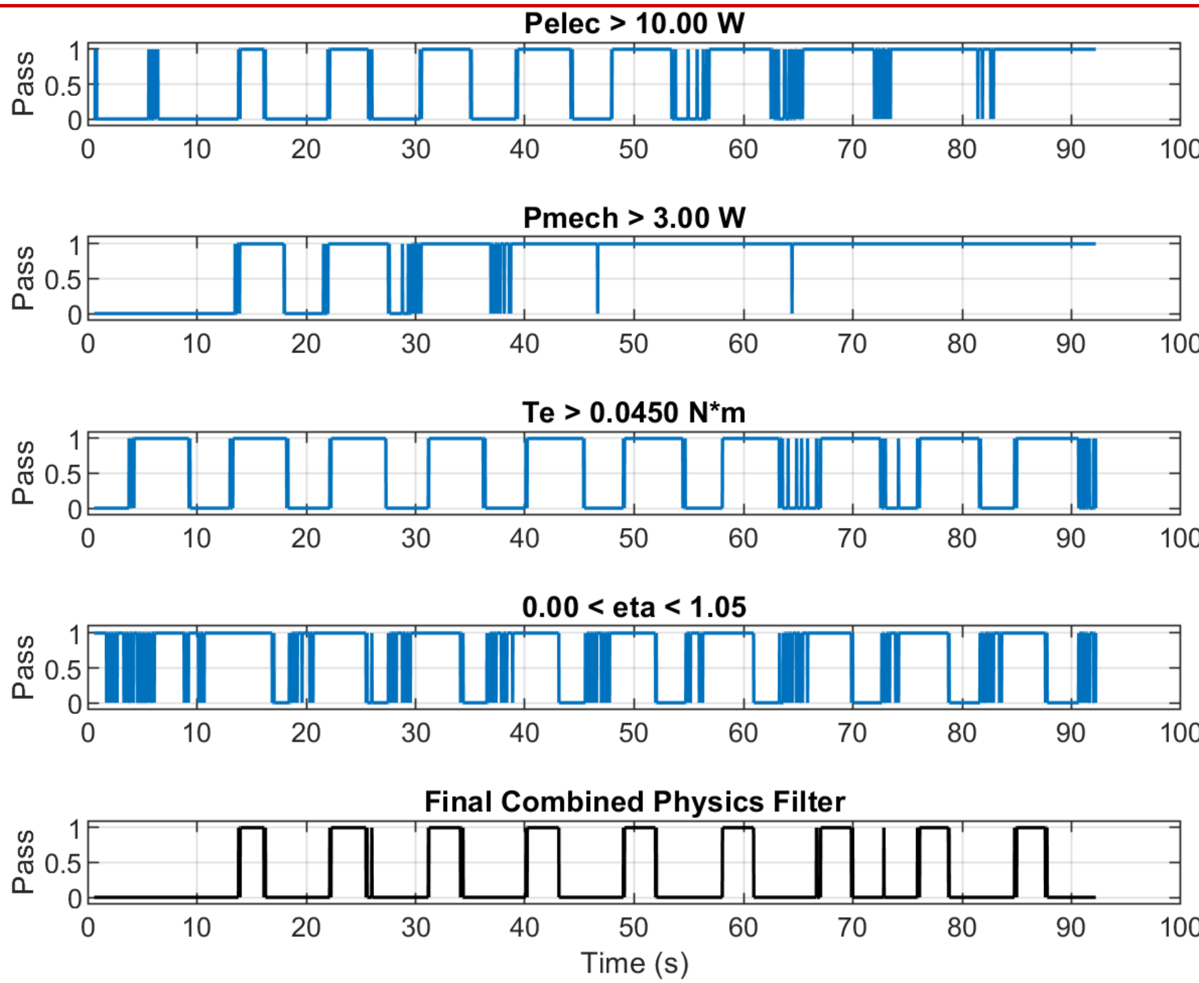

Figure 3. Screening masks used to separate valid operating points.

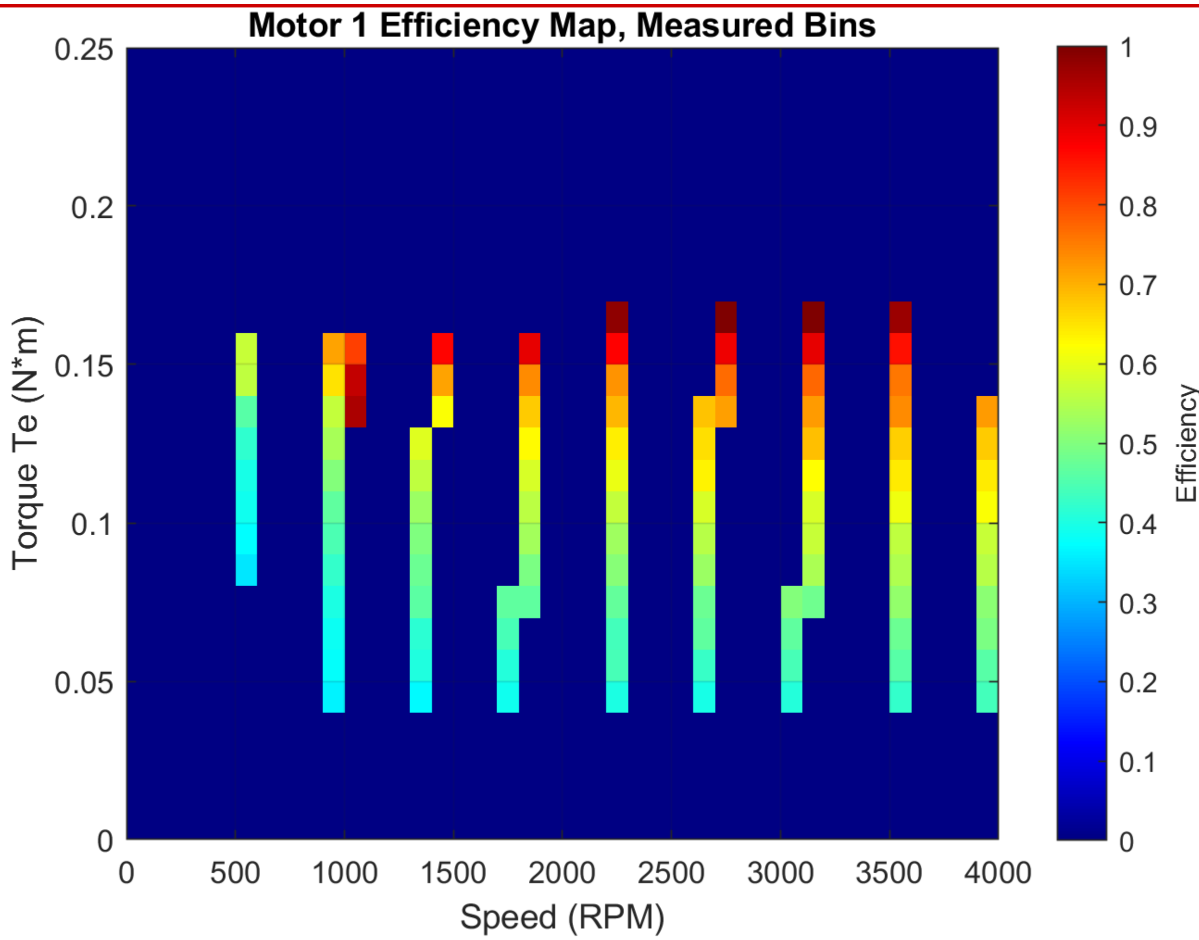

Figure 4. Measured-bin efficiency map; measured evidence is kept separate from interpolation.

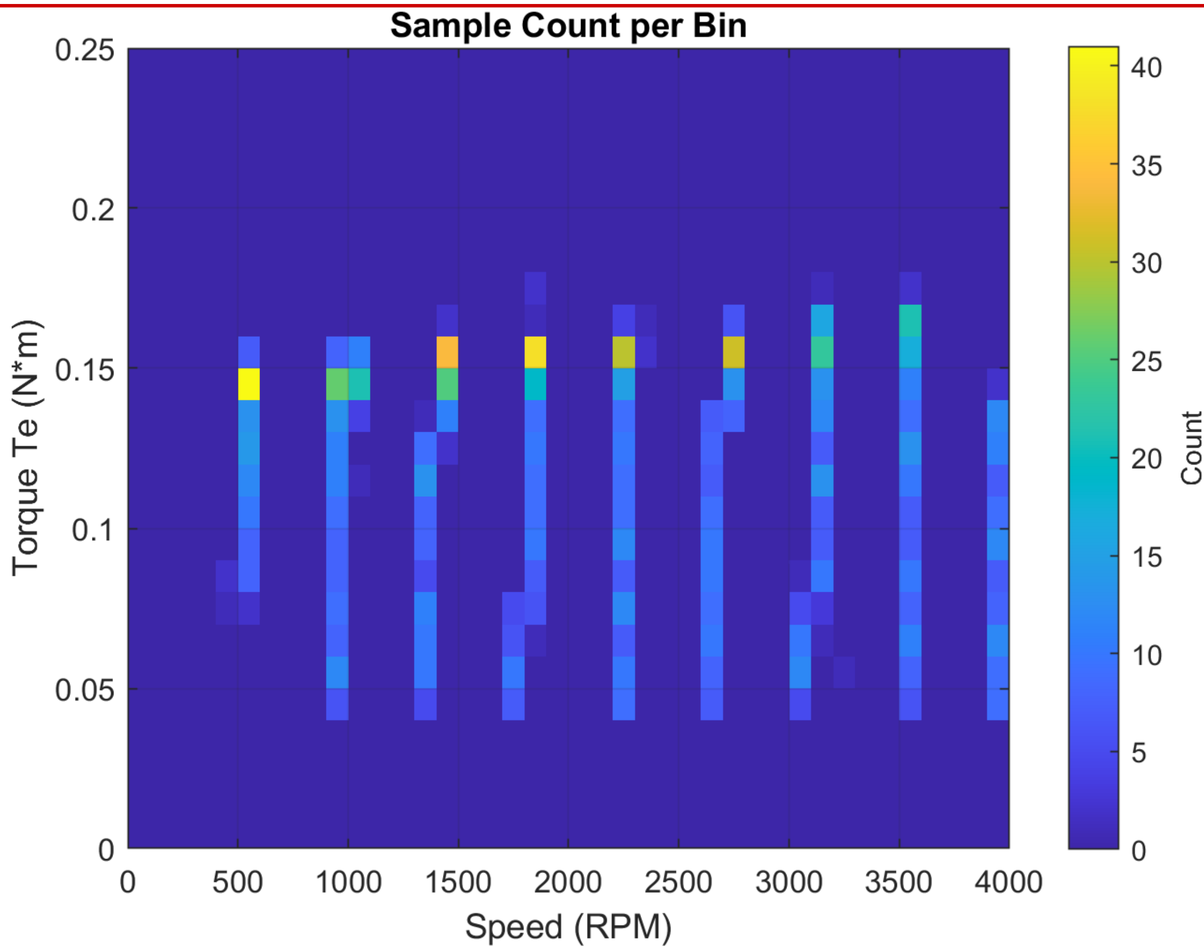

Figure 5. Sample count per operating bin for evidence density review.

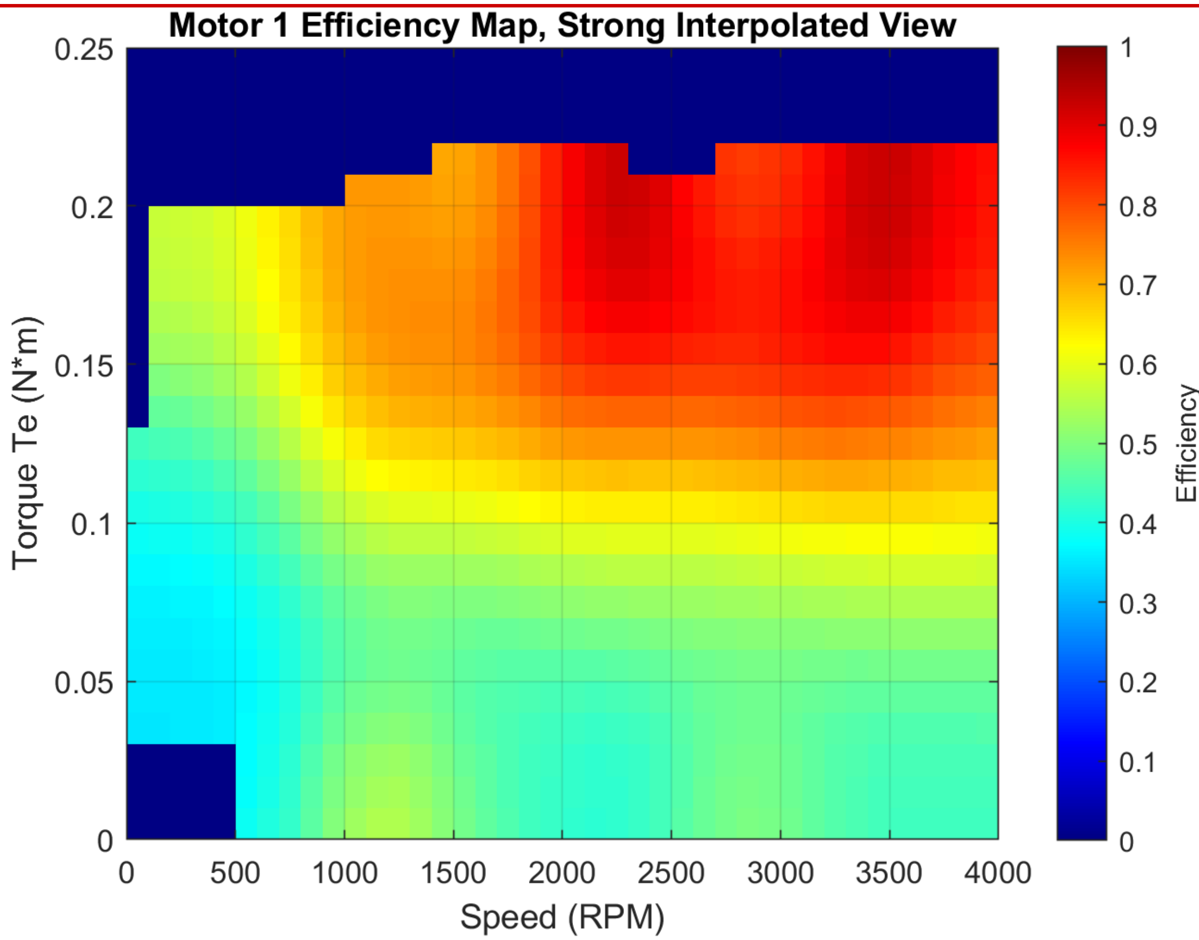

Figure 6. Interpolated efficiency surface for visualization only.

7. Requirements Verification Matrix

Requirement	Method	Result	Objective evidence
SYS-001	D/T	Verified	Operational dual-motor bench evidence.
SYS-002	I/T	Verified	Corrected phase and QEP assignments.
CTL-001	I/D	Verified	Target FOC/PWM/ADC/QEP architecture.
CTL-002	T	Verified	Successful speed-command tests and efficiency-map runs.
CTL-003	T	Verified	Fixed-dwell and drive-cycle load profiles.
CTL-004	I/D	Verified	Retained host/target model behavior and command routing.
COM-001	I/D	Verified	Generated target code mailbox configuration.
COM-002	I/D	Verified	Generated target code mailbox configuration.
COM-003	I/D	Verified	Byte pack/unpack and host command evidence.
COM-004	I	Verified	Interface summary and FRD architecture.
DAQ-001	I/A	Verified	Interpolation and crop logic in processing source.

Requirement	Method	Result	Objective evidence
DAQ-002	I/A	Verified	Validity masks and filtering plots.
DAQ-003	A	Verified	Sample-count map.
CHR-001	I/A	Verified	Processing edges and published figures.
CHR-002	I/A	Verified	Processing script and report method.
CHR-003	I/A	Verified	Processing script and report method.
CHR-004	I	Verified	Separate measured and interpolated maps.
DRV-001	I/D	Verified	Public drive-cycle setup script.
SAF-001	I	Verified	NOTICE and report limitations.
REL-001	I	Verified	Final package inspection.
PER-001	I/A	Verified	Repository or processing evidence.
PER-002	I/A	Verified	Repository or processing evidence.
PER-003	I/A	Verified	Repository or processing evidence.
TIM-001	D	Verified	Repository or processing evidence.
USA-001	I	Verified	Repository or processing evidence.
MAI-001	I	Verified	Repository or processing evidence.
TRC-001	I	Verified	Repository or processing evidence.
SEC-001	I	Verified	Repository or processing evidence.
SYS-021	I/T	Verified	Custom source configures SOC12/13 for ADCINA2/B2 with ADCINT5 and SOC14/15 for ADCINA1/B1 with ADCINT9.
PER-009	I	Verified	Both added pairs use ePWM1 trigger selection, simultaneous sampling, and a seven-clock acquisition window.

8. Validation Against Intended Use

Intended use	Validation statement	Result
Embedded motor-control bring-up	The platform demonstrates correction of physical interfaces and restoration of dual-machine operation.	Demonstrated
Real-time host integration	The platform demonstrates command/response transport between a real-time host and C2000 target over CAN.	Demonstrated
Dynamometer characterization	The platform demonstrates repeatable command generation, data screening, measured-bin mapping, and evidence-density review.	Demonstrated
Production motor-drive qualification	The public evidence does not include EMC, environmental, functional-safety, endurance, or production acceptance testing.	Not claimed
Calibrated metrology system	The release does not provide a complete uncertainty budget or traceable calibration chain.	Not claimed

9. Limitations, Risks, and Open Items

Item	Controlled statement
Safety	No independent safety controller, certified stop chain, or functional-safety assessment is claimed in this release.
Measurement uncertainty	Efficiency results are engineering characterization outputs; a full sensor, torque, timing, and interpolation uncertainty budget is not included.
CAN robustness	The public evidence establishes frame transport, not exhaustive bus-load, fault-injection, timeout, or recovery qualification.
Operating envelope	The analysis grid extends to 4000 rpm and 0.25 N m; this is not a universal continuous-duty rating for every hardware configuration.
Drive cycle	The compressed FTP-style profile is a bench demonstration and is not a regulatory vehicle-emissions test.
Third-party documentation	Vendor schematics are references and do not grant manufacturing or redistribution rights beyond their original terms.

10. Public/Private Release Boundary

Public evidence	Controlled or excluded
Architecture and requirement descriptions	Complete target and host models
Selected MATLAB setup/processing utilities	Generated C/C++ trees and binaries
Sanitized result plots	Raw unpublished datasets and tuning history
Vendor schematic PDFs and interface notes	Editable vendor CAD, Gerbers, pick-and-place, fabrication packages
Public BOM and repository map	Internal minutes, university paths, sponsor records, and design history
Custom F28069M ADC configuration source	TI device-support source and generated ADC build output

11. References and Document Control

- ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering (confirmed current by ISO in 2024).
- NASA Systems Engineering Handbook, Appendix C (good requirement guidance), Appendix D (requirements verification matrix), and Appendix E (validation matrix).
- Texas Instruments, LAUNCHXL-F28069M overview and TMS320F2806x real-time microcontroller documentation.
- Public repository evidence retained with this release.

Rev	Date	Author	Description
1.4	2026-08-04	HES	Final public release: added custom F28069M ADC SOC/interrupt configuration, actual editable Simulink models, and organized embedded/host source boundaries.

Appendix A - Board Interface Figure

The following retained board-interface figure documents the BoosterPack-to-LaunchPad signal boundary used during electrical-interface review. It is included as supporting public evidence and does not replace the current vendor schematic or board-revision documentation.

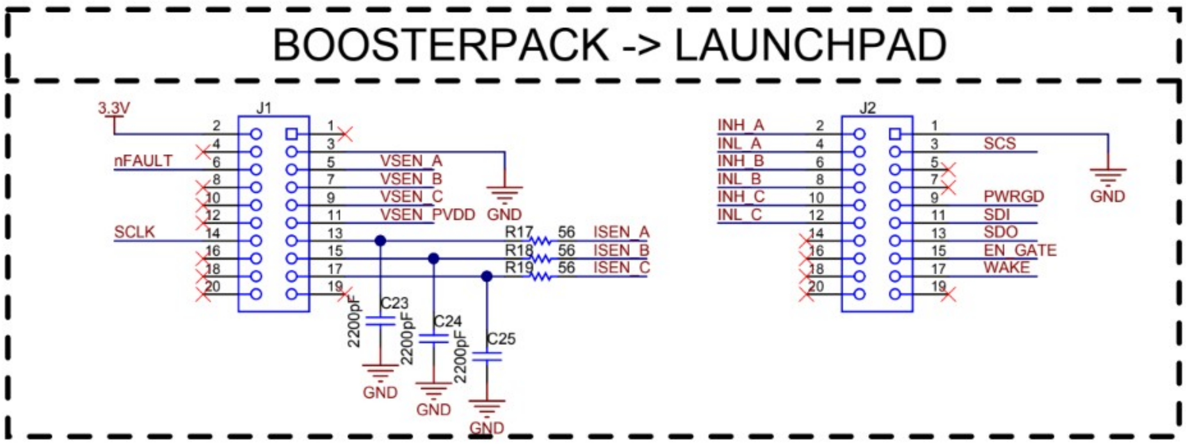

Figure A-1. BoosterPack-to-LaunchPad interface showing voltage-sense, current-sense, fault, SPI, power-good, shutdown, and control signal routing.

Use limitation: verify connector numbering, signal polarity, and board revision against current Texas Instruments documentation before energizing the platform.

Appendix B - Custom F28069M ADC Peripheral Integration

The public source release includes project-specific low-level ADC configuration in addition to the editable Simulink models. The retained file, F28069M_ADC_Custom_Config.c, directly configures F28069M ADC registers and is intentionally distinguished from MathWorks-generated MW_c28xx_adc.c output.

B.1 Functional purpose

The custom source extends the target acquisition path with two additional simultaneous analog-input pairs synchronized to ePWM1. This enables the embedded controller to acquire additional instrumentation channels without moving sample timing into the non-real-time host.

SOC pair	Inputs	Trigger	Completion	Role
SOC0/SOC1	ADCINA0 / ADCINB0	ePWM1	EOC1 -> ADCINT1	Baseline synchronized pair
SOC2/SOC3	ADCINA3 / ADCINB3	ePWM4	EOC3 -> ADCINT2	Baseline synchronized pair
SOC4/SOC5	ADCINA0 / ADCINB0	Software	Polled / model-controlled	Software-triggered pair
SOC6/SOC7	ADCINA3 / ADCINB3	Software	Polled / model-controlled	Software-triggered pair
SOC12/SOC13	ADCINA2 / ADCINB2	ePWM1	EOC13 -> ADCINT5	Added instrumentation pair
SOC14/SOC15	ADCINA1 / ADCINB1	ePWM1	EOC15 -> ADCINT9	Added instrumentation pair

B.2 Implementation details

- SOC12/SOC13 select ADCINA2 and ADCINB2 in simultaneous-sampling mode, use ePWM1 as the trigger, and route EOC13 to ADCINT5.
- SOC14/SOC15 select ADCINA1 and ADCINB1 in simultaneous-sampling mode, use ePWM1 as the trigger, and route EOC15 to ADCINT9.
- The acquisition window is configured as ACQPS = 6, corresponding to seven ADC clock cycles.
- Late interrupt pulse positioning is enabled so the ADC result latch is updated before the interrupt is asserted.
- The file retains baseline PWM-triggered and software-triggered ADC pair configuration used by the dual-motor target.

B.3 Verification and ownership boundary

Verification is by source inspection and target integration test. The public repository includes the project-specific configuration file and editable target model, but excludes TI device-support implementation, generated code, object files, and binaries. This demonstrates custom peripheral integration without claiming authorship of vendor libraries or generated support code.